

The arc rig, after the loudspeaker campaign

Eleven UMIK-2 capsules on a 1.68 m arc · what is the instrument, what is the room, and what the source can be trusted for · rev 1, 2026-09-17 · every number recomputed from our own captures

The arc is a good instrument and the map is mostly the room. Of the per-position error, 0.21 dB rms is provably the rig and 1.04 dB rms is provably the room; the 0.84 dB that a single flip could not divide is, on a second and independent test, essentially all room. The arc repeats to 0.026 dB when nothing is touched. The hub loudspeaker is a valid axisymmetric source below 3 kHz and unusable from 3–5 kHz.

HOW IT WAS MEASURED

- Four ideas, each one a difference that cancels what we cannot model.
- **Capsules first (16 Sept).** Every capsule against every other at one seat, 55 pairs. Differencing two capsules in the same place cancels source, room and analog chain *exactly*; the source need only be stationary, and it was (reference repeated to -0.025 dB over 2 h 13 min). Seven of eleven factory files were wrong by 0.6–3.6 dB. Corrections folded into `cal.gain_db`; spread across the eleven fell 4.01 → 0.02 dB.
 - **The whole arc at once (17 Sept).** A 1 l printed sphere at the ring centre firing along the symmetry axis, so all eleven sit at the same polar angle and source directivity cancels. 112 tones, 257–6350 Hz, 24/octave, all capsules recorded simultaneously.
 - **Flip the arc.** What travels with the arc keeps its label; what stays in the room swaps between mirrored labels. Un-flip the second run before averaging, and each half shows its own subject whole.
 - **Stand it vertical.** A completely different room geometry with the same hardware — the only way to break the tie a flip leaves behind.

WHAT THE NUMBERS SAY

QUANTITY	VALUE
Run-to-run repeatability, nothing touched	0.026 dB
Worst single cell, same	0.20 dB
Position spread, arc horizontal	3.86 dB
Position spread, arc vertical	4.76 dB
Provably the rig (antisymmetric under the flip)	0.21 dB
Provably the room (antisymmetric, un-flipped)	1.04 dB
Shared symmetric term, undividable by one flip	0.84 dB
...resolved by the vertical session: common part	≤ 0.16 dB

The reorientation is what settles it. A term that travels with the arc must appear in both geometries; the horizontal and vertical symmetric parts share $cov = -0.09 \text{ dB}^2$, 95 % CI [-0.22, +0.03] by frequency-block bootstrap — consistent with zero. So the shared 0.84 dB is the room, and only ~0.2 dB of the position map is arc hardware.

THE SOURCE HAS A BAD BAND 3–5 KHZ

The method needs one property of the loudspeaker: that its radiation be **rotationally symmetric about its aim** — every capsule sits at 90° to that axis. It need not be omnidirectional. Rolling such a source about its own axis is a no-op *by construction*, which makes the roll a clean test.

Rotating the **whole sphere** gave a false answer: putting it back at the same roll reproduced only to 1.32 dB, so handling the enclosure costs ~1.3 dB on its own — 56 % of the effect being claimed. The honest test is to move the part under test: **unbolt the driver and remount it 180° inside a sphere that never moves.**

DRIVER TURNED 180°, ENCLOSURE FIXED	CHANGE
257 Hz – 3 kHz (86 tones, worst 0.58)	0.20 dB
3–4 kHz	1.18 dB
4–5 kHz	5.12 dB
4362 Hz alone ($k_a = 2.39$)	11.0 dB
5–6.35 kHz	0.71 dB

At 4362 Hz the arc pattern does not shift or scramble — it **turns over** (-90°: +3.8 → -13.0 dB; +90°: -9.3 → +8.0; $r = -0.63$). A 180° rotation inverting the pattern is an **m = 1 rocking mode**: the cone pivoting about a diameter instead of pistoning, its axis bolted to the driver. It is the cone, so a different enclosure would not help. Enforced in code as `calibrator.rig.SOURCE_SUSPECT_HZ`.

WHAT THIS CHANGES FOR MEASUREMENTS

- **A position map below 3 kHz is trustworthy** to ~0.2 dB of rig error. Above 3 kHz, exclude the source band or use a different source.
- **The room, not the rig, is what to treat.** Effort belongs on the surfaces and the source position, not on rebuilding clamps.
- **The loudspeaker map does not reproduce the propeller map** ($r = -0.04$ tone-notched, +0.04 mixed; all 22 relabellings tested, best +0.24). So the September pattern is not simply "the room as any source sees it" — the propeller's own inflow, or the object at the hub, still matters. **This is the open question.**
- **Identity is manual.** Every UMIK reports USB serial 00000; capsule-to-position was confirmed by reading labels, and a remount can mirror the map without changing a cable — handled by `labels_mirrored` in each run's `meta.json`.

STILL OPEN

- Why the loudspeaker and the propeller disagree — inflow asymmetry, or the hub object.
- The **absolute** anchor: substitution calibration is blind to error common to all eleven. Needs a 94 dB calibrator.
- Room geometry is still unrecorded: hub height, room L×W×H, foam, what stands near the arc ends.
- A repeat run in the vertical configuration, for that geometry's own noise floor.

Data and code. Sessions in `calibrator/sessions/2026-09-16` (capsules) and `2026-09-17` (arc); levels and meta mirrored to the measurement repo, WAVs local. `calibrator/rig.py` captures and `read_map()` is the single loader, so a figure cannot disagree with the capture about what a level means. Figures: `flip-split.png`, `driver-rotation.png`, `source-tests-quad.png`. **This page supersedes** `arc-validation-onepager`, `arc-validation-remedies`, `arc-validation-guide` and `hub-source-onepager`, which were written while the loudspeaker session was still future work and describe it as the thing that "will decide". Where this page and `CLAUDE.md` disagree, `CLAUDE.md` governs — it is the record these numbers were audited into.